

Question number	Answer	Additional guidance	Mark												
3(a)	<table><tr><td></td><td></td><td></td></tr><tr><td>P</td><td>medulla oblongata</td><td>non-conscious regulation of body processes / example of process regulated by medulla</td></tr><tr><td>Q</td><td>hypothalamus</td><td>homeostasis / thermoregulation</td></tr><tr><td>R</td><td>cerebellum</td><td>balance and coordinating movement</td></tr></table>				P	medulla oblongata	non-conscious regulation of body processes / example of process regulated by medulla	Q	hypothalamus	homeostasis / thermoregulation	R	cerebellum	balance and coordinating movement	One mark for two correct cells Accept osmoregulation	(2)
P	medulla oblongata	non-conscious regulation of body processes / example of process regulated by medulla													
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Question number	Answer	Additional guidance	Mark
3(b)(i)	C CT scans produce a cross sectional image of a thin slice through the body		(1)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A description that includes three of the following points: • isotope {that emits positrons / is incorporated into glucose} (1) • more active neurones with increased respiration (1) • will require increased supply of glucose (1) • positrons (emitted from glucose) produce gamma rays that detected and are converted into an image (1)	Accept suitable named isotope e.g. carbon-11 or oxygen-15, radioactive material in glucose Accept will {absorb/use} more glucose	(3)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes the following points: • {impulses / electrical activity} from the neurones at site of seizure (1) • {arrive at / travel to} the thermoregulatory centre in the hypothalamus (1) • description of consequence that would lead to increased body temperature (1)	ACCEPT impulses travel to sweat glands e.g. reduce sweating / increase metabolic rate increased respiration	(3)

